THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET A

Chapter 8: **Predicting What Comes Next: Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. For the sequence $t_n = 3n - 4$, the value of t_{12} is: 1

(a) 33 **(b)** 31 **(c)** 32 **(d)** 64

2. For the sequence $t_n = 4n + 1$, the value of t_{20} is: 1

(a) 82 **(b)** 80 **(c)** 162 **(d)** 81

3. Which term of the sequence $t_n = 3n - 7$ is 332? 1

(a) $n = 115$ **(b)** $n = 114$ **(c)** $n = 113$ **(d)** $n = 112$

4. The 15th term of the AP $21, 18, 15, \ldots$ is: 1

(a) −21 **(b)** −22 **(c)** −20 **(d)** −42

5. The 11th term of the AP $7, 13, 19, \ldots$ is: 1

(a) 67 **(b)** 68 **(c)** 134 **(d)** 66

6. The sum of the first 100 natural numbers is: 1

(a) 5050 **(b)** 10100 **(c)** 5049 **(d)** 5051

7. The 7th term of the GP with first term 5 and common ratio 2 is: 1

(a) 319 **(b)** 640 **(c)** 320 **(d)** 321

8. If $s_1 = 3$ and $s_n = s_{n-1}(s_{n-1} - 1)$, then s_3 is: 1

(a) 6 **(b)** 30 **(c)** 870 **(d)** 18

9. The sequence $3, 6, 12, 24, \ldots$ is: 1

(a) a GP with $r = 3$ **(b)** a GP with $r = 2$

(c) neither an AP nor a GP **(d)** an AP with $d = 3$

10. The sum of the first n odd numbers is: 1

(a) $\frac{n(n+1)}{2}$ **(b)** $2n$ **(c)** n^2 **(d)** $n(n+1)$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the first five terms of the sequence $t_n = 3n - 4$. 2

12. Find the first five terms of $t_n = 2 - 5n$. 2

13. Find the first five terms of $t_n = n^2 - 2n + 3$. 2

14. Find the 10th and 15th terms of $t_n = 5n - 3$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Which term of $t_n = 3n - 7$ is 332? Is 557 a term of this sequence? 3

16. Check whether 308 and 471 are terms of $s_n = 5n - 2$. 3

17. An AP has 50 terms; its 3rd term is 12 and its last term is 106. Find the 29th term. 3

OR

How many two-digit numbers are divisible by 3? Find their sum. 3

18. Harish starts at ₹5,00,000 a year and gets a raise of ₹20,000 each year. After how many raises does his salary reach ₹6,00,000? 3

OR

A child arranges marbles in rows: 1 in the first row, 2 in the second, … up to 25 rows. How many marbles are used? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Prove that $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$. 5

(ii) Hence find S_{20} and S_{50}.

(iii) Find $25 + 26 + \cdots + 58$.

OR

Case study: A taxi company charges a fixed booking fee of ₹200 plus ₹40 per km. 5

(i) Write the fares for 1, 2 and 3 km.

(ii) Show the sequence is an AP and find the nth term.

(iii) Find the fare for 10 km.

(iv) For how many km is the fare ₹1,000?

20. **Case study:** A child arranges marbles in rows of 1, 2, 3, … 5

(i) How many marbles are in the first 10 rows?

(ii) How many in 25 rows?

(iii) How many rows are needed for exactly 210 marbles?

(iv) Which triangular number is the answer to (i)?

OR

Case study (Sierpiński triangle): Stage 0 is a black triangle of area 1 sq. unit. 5

(i) Give the number of black triangles at Stages 0 to 3.

(ii) Give the number at Stages 4 and 5.

(iii) Write the rule for Stage n.

(iv) Find the black area at Stage 3 and at Stage n.

(v) What happens to the area as n grows?

END OF QUESTION PAPER — Total: 40 Marks